

Multiple Choice

1. **Answer:** D

Options: A) 15, 17; B) 15, 19; C) 17, 19; D) 17, 21

Note: Correct option: D - 17, 21

2. **Answer:** B

Options: A) $1850x$; B) $2000 - 150x$; C) $150x$; D) $2000 + 150x$

Note: Correct option: B - $2000 - 150x$

3. **Answer:** B

Options: A) $3k+1$; B) $5k$; C) $3k$; D) 1

Note: Correct option: B - $5k$

4. **Answer:** B

Options: A) add 6 to both sides; B) divide both sides by 6; C) multiply both sides by 6; D) subtract 6 from both sides

Note: Correct option: B - divide both sides by 6

5. **Answer:** D

Options: A) 11.9 in.; B) 11 in.; C) 45.0 in.; D) 12 in.

Note: Correct option: D - 12 in.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '\$5.52'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '46 over 108'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Let s be the side length of the square. Then the dimensions of each rectangle are $s \times \frac{s}{5}$. The perimeter of one of the rectangles is $s + \frac{s}{5} + s + \frac{s}{5} = \frac{12}{5}s$. Setting $\frac{12}{5}s = 36$ inches, we find $s = 15$ inches. The perimeter of the square is $4s = 4(15in.) = 60$ inches.
Note: Students should show all steps in their mathematical solution.
12. **Answer:** -9. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.